Umberto Domenico Ciccia

Site Reliability Engineer, Software Oriented | Cloud Infrastructure | Distributed Systems & Automation

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Experience

Amazon Web Services (AWS), Dublin, Ireland, Nov 2025 – present
System Development Engineer

- Redesigned DNS infrastructure following root-cause analysis of a recurring misconfiguration, eliminating the underlying issue and reducing SEV-2 alerts from 84 to zero over six months across all AWS partitions.
- Led the refactoring of 184 Route 53 infrastructure pipelines, increasing the percentage with artifacts less than 40 days old from 80% to 98% in eusc-de-east-1, improving infrastructure software and patch freshness.
- Unblocked a strategic customer launch under C-level executive escalation by delivering DNSSEC support for Route 53 in AWS European Sovereign Cloud 3 weeks ahead of schedule, enabling migration without changes to the customer's existing infrastructure.
- Delivered the Route 53 CIDR routing feature end-to-end in the European Sovereign Cloud 4 months ahead of schedule, implementing infrastructure, APIs, and console support for customer-configurable IP-based DNS routing.
- Maintained operational reliability of Route 53, AWS WAF, and AWS Shield in AWS European Sovereign Cloud through production incident response, on-call rotations, and root-cause-driven improvements, supporting services with demanding availability commitments.
- Authored incident postmortems and operational runbooks to document root causes, standardize troubleshooting procedures, and improve incident response and service reliability.

NTT Data, Cosenza, Italy, Jan 2025 – Oct 2025
Cloud Software Engineer

- Replaced manual Bash-script deployments with Jenkins CI/CD pipelines for containerized applications on Amazon EKS, automating the release and deployment process for ItaliaOnline.
- Designed and deployed a multi-cloud infrastructure spanning AWS and on-premises using Terraform, Kubernetes, EKS, Aurora, DMS, S3, and ECR, reducing deployment failures by 70% and improving deployment consistency for ItaliaOnline
- Built GCP data ingestion pipelines processing 10k+ daily records, improving data availability and release reliability using Dataflow, Compute Engine, and Cloud Storage for Coop Italia
- Built an AI-powered Quarkus agent using the Vertex AI SDK to analyze user-defined multi-cloud architectures and recommend optimized configurations, reducing estimated infrastructure costs by 30% versus the original designs on the NTT KUMO platform.

Vision One, Recanati, Italy, July 2024 – Dec 2024
Software Engineer

- Developed an on-call rotation scheduling system using .NET and Blazor, automating operator-to-shift assignments and improving allocation efficiency by 80% compared with the previous manual process.
- Refactored legacy code, removing 60% of unused code to simplify the codebase and improve maintainability.
- Introduced automated testing with xUnit, achieving 80% test coverage to strengthen regression detection and release confidence.

Caliò Informatica, Cosenza, Italy, Jan 2024 – July 2024
Software Engineer

- Developed workflow-building capabilities for AIDA, an OpenAI-powered automation platform, enabling non-technical users to create and configure AI workflows through a visual drag-and-drop interface.
- Optimized OpenAI API requests and prompts, reducing costs by 80% and improving response performance by 60%.

Skills

SRE & Reliability Engineering: SLIs/SLOs, Error Budgets, Incident Management, Root Cause Analysis (RCA), On-Call Operations, Capacity Planning, Performance Engineering, Resilience & Disaster Recovery

Cloud & Infrastructure: AWS, GCP, Azure, Linux, TCP/IP, HTTP/TLS, DNS, Kubernetes, Docker, Helm, Terraform, Pulumi, Crossplane, AWS CDK, Ansible

Observability: Prometheus, Grafana, Monitoring, Alerting, Logging, Distributed Tracing

CI/CD & Automation: Jenkins, Argo CD, Git, Bash, Python, PowerShell, Safe Deployments, Change Management, Infrastructure Automation

Programming Languages, Frameworks & Databases: Java, TypeScript, JavaScript, C#, SQL, C++, Quarkus, Spring Boot, .NET, FastAPI, PostgreSQL, SQL Server, MongoDB, Redis, DynamoDB

Languages: Italian (Native), English (Professional)

Education

Università della Calabria, M.Sc. in Artificial Intelligence and Machine Learning – GPA: 4.0/4.0

- Relevant coursework: Artificial Intelligence, Machine Learning, Computer Vision, Distributed System, Knowledge representation and reasoning, Statistical Learning

Università della Calabria, B.Sc. in Computer Engineering – GPA: 4.0/4.0

- Relevant coursework: Operating Systems, Computer Networks, Database Systems, Object Oriented Programming, Software Engineering